PREFACE

In the contemporary landscape of retail marketing, the need for data-driven decision-making has never been more critical. FOOTWEAR FIRM, a globally recognized brand in the footwear and fashion accessory sector, faces the complex challenge of optimizing its marketing strategies to maximize return on investment (ROI) and drive growth. Traditional methods of gauging campaign performance often fall short in capturing the nuanced impacts of various media channels and external factors on sales. Hence, there is a pressing need for a sophisticated analytical approach that can provide deeper insights into marketing effectiveness.

This document presents a comprehensive study on Market Mix Modelling (MMM) specifically designed for FOOTWEAR FIRM. The primary objective of this project is to develop an advanced MMM framework that not only evaluates the performance of FOOTWEAR FIRM's 360-degree and regional marketing campaigns but also identifies the most effective media channels for increasing sales and revenue. By leveraging key metrics such as Return on Ad Spend (RoAS) and Marginal RoAS, this model aims to provide actionable insights into both current and potential future returns on marketing investments.

The study employs a holistic approach by integrating a wide range of data sources, including internal sales data, shop and item master data, campaign calendars, and external factors like economic indicators and seasonal trends. This integration ensures that the model accurately reflects the multifaceted influences on consumer purchasing behaviour. Through meticulous data pre-processing and feature engineering, the study prepares a robust foundation for model building.

Utilizing PySTAN, a powerful tool for Bayesian optimization, the study constructs models at the regional level to account for the unique market dynamics across different areas. The models incorporate various transformations and optimizations to identify the most impactful media spends and suggest strategic reallocations to enhance overall ROI.

The results provide a detailed analysis of the contributions of different media channels and campaigns to sales, offering insights that are crucial for making informed marketing decisions. The study also explores the impact of specific campaigns, such as those targeting sneakers, to validate the model's effectiveness and align promotional strategies with actual sales trends.